

Pathway-Aware Multimodal Transformer for Explainable Survival Prediction in Bladder Cancer

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ABSTRACT – Cancer survival prediction requires effective integration of heterogeneous biomedical modalities such as gene expression profiles and histopathological whole-slide images (WSIs) to capture both molecular and morphological characteristics of tumors. Existing approaches often rely on unimodal learning or coarse multimodal fusion, limiting their ability to model fine-grained genotype-phenotype interactions. This paper presents a survey and system-oriented analysis of a pathway-aware multimodal framework for bladder urothelial carcinoma (BLCA) survival prediction based on the Pathway-Aware Multimodal Transformer (PAMT). The framework organizes gene expression data into biologically meaningful pathway representations while processing WSIs using patch-based feature extraction and transformer-based encoding. A contrastive alignment mechanism and pathway-to-patch cross-attention fusion module enable biologically guided multimodal interaction learning between genomic pathways and localized tissue regions. The integrated representation is utilized for survival risk estimation using a Cox proportional hazards-based objective, while attention-based interpretability mechanisms provide insights into prognostically significant pathways and tissue structures. The survey findings indicate that pathway-aware transformer architectures offer improved multimodal interaction modeling, enhanced interpretability, and strong potential for clinically explainable cancer survival prediction systems.

KEY WORDS: Multimodal Learning, Survival Prediction, Bladder Cancer, Transformer Architecture, Explainable AI

1. INTRODUCTION

Cancer survival prediction plays a critical role in clinical decision-making, enabling risk stratification and personalized treatment planning. In bladder urothelial carcinoma (BLCA), patient outcomes are influenced by complex interactions between molecular-level genetic activity and observable tissue-level morphology. Gene expression data provides insight into biological processes driving tumor progression, while histopathological whole-slide images (WSIs) capture structural and cellular characteristics of tumor tissue. However, analyzing

these modalities independently limits the ability to understand the complete disease landscape.

Traditional survival prediction approaches primarily rely on single-modality data, such as gene expression or histopathology images. While gene-based models capture molecular signals, they fail to reflect spatial tumor heterogeneity. Conversely, image-based models capture morphological patterns but lack direct information about underlying biological mechanisms. To address these limitations, multimodal learning approaches have been introduced to integrate genomic and imaging data. However, most existing methods perform coarse-level feature fusion, treating each modality as a single global representation. This approach ignores fine-grained interactions between specific biological pathways and localized tissue regions, which are essential for understanding cancer progression.

Recent advances in transformer-based architectures provide a powerful framework for modeling complex relationships within and across modalities. In particular, pathway-aware representations enable the incorporation of biological prior knowledge by organizing gene expression data into functional groups, while patch-based processing allows scalable analysis of high-resolution WSIs. These developments create an opportunity to model fine-grained interactions between genotype and phenotype in a structured and interpretable manner.

In this paper, we present a survey and system-oriented study of a pathway-aware multimodal framework for survival prediction in bladder cancer, based on the Pathway-Aware Multimodal Transformer (PAMT). The approach integrates pathway-level genomic representations with patch-level image features using transformer-based encoding, contrastive alignment, and cross-attention fusion. In addition to predictive performance, the framework provides interpretability by identifying important biological pathways and highlighting relevant regions in histopathology images.

The main contributions of this work are as follows:

1. A structured analysis of multimodal survival prediction approaches in cancer.

2. A pathway-aware representation strategy for integrating gene expression and histopathology data.
3. A fine-grained multimodal interaction framework using alignment and cross-attention mechanisms.
4. An interpretable prediction pipeline that links biological pathways to spatial tissue features.

2. METHODS

The methodological framework adopted in this survey follows a structured literature review approach aimed at analyzing recent advancements in multimodal cancer survival prediction systems, with particular focus on transformer-based architectures integrating genomic and histopathological data. The survey investigates how artificial intelligence techniques, especially pathway-aware multimodal learning frameworks, improve survival prediction accuracy and interpretability in computational oncology.

The primary objective of this survey is to examine existing approaches for integrating gene expression data, whole-slide histopathology images, pathway-aware representations, attention mechanisms, and survival analysis models within a unified predictive framework. Special emphasis is placed on transformer-based multimodal systems such as Pathway-Aware Multimodal Transformer (PAMT) and related deep learning architectures developed for interpretable cancer prognosis prediction.

2.1 Search Strategy

The literature search was conducted using major scientific databases including IEEE Xplore, Google Scholar, ScienceDirect, PubMed, and ACM Digital Library. Keywords used during the search process included combinations of:

- “multimodal cancer survival prediction”
- “pathology and genomics fusion”
- “transformer for computational pathology”
- “whole slide image survival analysis”
- “pathway-aware deep learning”
- “multimodal transformer in healthcare”
- “cancer prognosis prediction using AI”
- “PAMT architecture”
- “survival analysis using deep learning”

The survey primarily focused on studies published between 2018 and 2025 to capture recent developments in transformer architectures, multimodal fusion techniques, and explainable medical AI systems.

2.2 Inclusion and Exclusion Criteria

The inclusion criteria required selected papers to address at least one of the following areas:

- Cancer survival prediction
- Multimodal integration of genomic and pathology data
- Transformer-based medical learning frameworks
- Pathway-aware neural architectures
- Attention-based fusion mechanisms
- Interpretable AI techniques in computational pathology

Studies focusing exclusively on unrelated biomedical domains, low-level image enhancement, hardware-specific optimization, or non-clinical bioinformatics tasks were excluded from the survey. Papers lacking methodological clarity, experimental validation, or reproducible evaluation metrics were also excluded from the final review.

2.3 Analysis Framework

The selected studies were comparatively analyzed using multiple evaluation criteria including:

- Model architecture
- Modality integration strategy
- Feature representation methods
- Survival prediction techniques
- Interpretability mechanisms
- Dataset utilization
- Evaluation metrics
- Computational complexity

Particular attention was given to how different studies addressed fine-grained genotype-phenotype interaction modeling, pathway-level biological representation, and cross-modal alignment between genomic and histopathological information.

Additionally, the survey examined limitations of existing approaches including insufficient interpretability, coarse multimodal fusion, computational inefficiency in whole-slide image processing, and limited biological contextualization. This comparative analysis was used to identify current research gaps and motivate the need for pathway-aware multimodal transformer frameworks for interpretable cancer survival analysis.

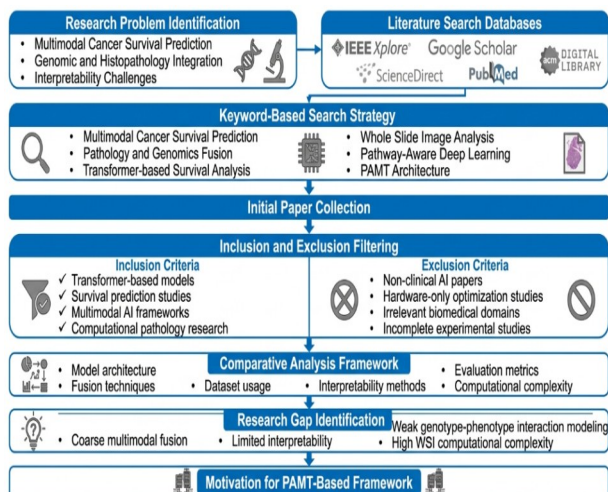


Fig. 1: Survey Methodology Flowchart

3. RESULTS

From the literature review, multiple research domains related to cancer survival prediction and multimodal artificial intelligence were identified. The surveyed studies primarily focused on genomic survival analysis, computational pathology, multimodal learning frameworks, and transformer-based architectures. The following sections summarize the major findings from each domain and highlight the limitations present in existing systems.

3.1 Genomic-Based Survival Prediction

Several studies focused on predicting cancer survival using genomic and transcriptomic information such as RNA-seq gene expression data. Traditional statistical methods including Cox proportional hazards regression and random survival forests were widely used for survival estimation. Recent deep learning approaches demonstrated improved capability in learning nonlinear relationships between genes and survival outcomes.

Although genomic models achieved promising predictive performance, many studies suffered from high-dimensional feature complexity and poor biological interpretability. Most approaches treated genes independently without incorporating pathway-level biological relationships.

3.2 Histopathology-Based Survival Prediction

Computational pathology has emerged as an important research area for cancer prognosis prediction using histopathological whole-slide images (WSIs). Convolutional Neural Networks (CNNs), Multiple Instance Learning (MIL), and transformer-based pathology models were commonly used for extracting prognostic tissue features.

Transformer-based pathology frameworks showed improved performance in modeling spatial tissue

relationships and tumor heterogeneity. However, most WSI-based systems require extensive preprocessing pipelines involving patch extraction, tissue segmentation, and high computational resources due to gigapixel image sizes.

3.3 Multimodal Cancer Survival Frameworks

Several recent studies explored multimodal frameworks integrating genomic data, histopathology images, and clinical information for survival prediction. These systems generally outperformed unimodal approaches because they captured complementary molecular and morphological characteristics of tumors.

Existing multimodal approaches utilized:

- Early fusion strategies
- Late fusion mechanisms
- Co-attention networks
- Graph-based learning
- Transformer-based cross-modal interaction

Despite improved performance, many studies performed only coarse-level modality fusion and failed to model fine-grained biological interactions between genomic pathways and tissue morphology.

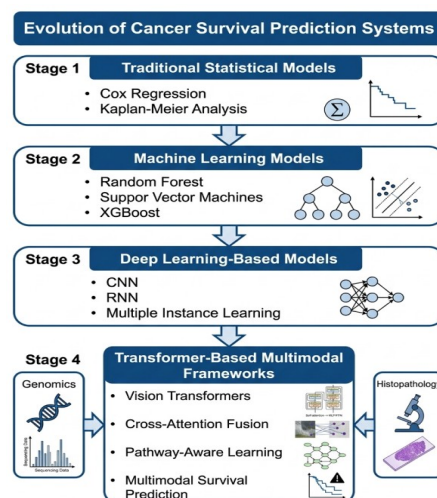


Fig. 2: Proposed Integrated System Architecture

3.4 Transformer-Based and Pathway-Aware Architectures

Transformer architectures have recently gained significant attention in computational oncology due to their ability to model long-range dependencies and multimodal interactions. Vision Transformers (ViTs), multimodal transformers, and attention-based fusion frameworks demonstrated improved survival prediction capability compared to conventional deep learning models.

Pathway-aware architectures further improved interpretability by organizing genes into biologically meaningful pathway representations instead of processing isolated gene features. Such approaches enhanced biological contextualization and explainability in survival prediction systems.

Among the surveyed approaches, the Pathway-Aware Multimodal Transformer (PAMT) framework demonstrated strong potential for integrating genomic pathways and histopathological representations using cross-attention interaction mechanisms.

Paper	Modalities	Fusion Method	Dataset	Interpretability	Limitation
Pathomic Fusion	Histology + Genomics	Fusion Network	TCGA	Medium	coarse fusion
MCAT	WSI + Omics	Co-Attention	TCGA	High	computational cost
AMMASurv	WSI + RNA-seq	Attention Fusion	TCGA	Medium	limited pathway modeling
PAMT	Pathology + Gene Expression	Cross-Attention	TCGA	High	computational complexity

Table. 1: Comparative Analysis Of Mutlimodal Cancer Survival Prediction Frameworks

3.5 Comparative Analysis and Identified Research Gaps

The comparative analysis revealed several important limitations in existing cancer survival prediction systems. Most approaches primarily focused on predictive accuracy while offering limited interpretability regarding the biological reasoning behind predictions.

Additionally, many multimodal systems lacked effective genotype-phenotype interaction modeling due to simplistic fusion strategies. Computational inefficiency associated with whole-slide image processing and inadequate cross-modal alignment mechanisms also remained major challenges.

The survey findings indicate that future cancer survival prediction systems require:

- Biologically interpretable architectures
- Pathway-aware genomic representation
- Efficient multimodal fusion
- Improved genotype-phenotype interaction modeling
- Scalable transformer-based frameworks

These observations motivate the development of pathway-aware multimodal transformer systems such as PAMT for interpretable cancer survival analysis.

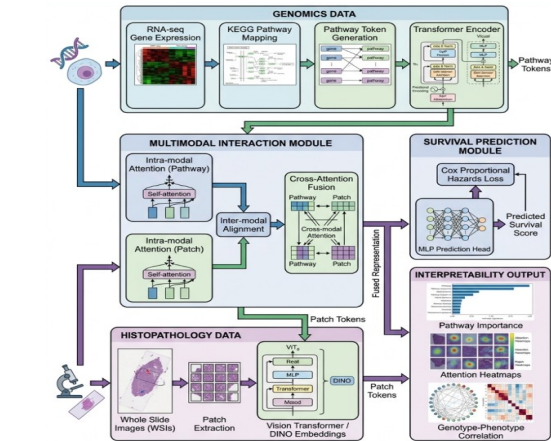


Fig. 2: Comparative Analysis and Research Gaps in Existing Cancer Survival Prediction System

4. DISCUSSION

The survey findings indicate that cancer survival prediction systems are rapidly evolving from traditional statistical approaches toward transformer-based multimodal artificial intelligence frameworks. The integration of genomic, histopathological, and clinical modalities has significantly improved the ability of AI systems to capture complex biological relationships associated with tumor progression and patient prognosis.

One of the major observations from the reviewed studies is the growing importance of multimodal learning in computational oncology. Unimodal systems based solely on genomic data or histopathological images often fail to capture the complete biological complexity of cancer. In contrast, multimodal frameworks leverage complementary molecular and morphological information, enabling more robust and clinically meaningful survival prediction.

Another important trend identified in the literature is the increasing adoption of transformer architectures for medical AI applications. Transformer-based systems demonstrated superior capability in modeling long-range dependencies, cross-modal interactions, and complex feature relationships compared to conventional convolutional neural networks and traditional machine learning approaches. Attention mechanisms also improve interpretability by identifying biologically relevant pathways and tissue regions contributing to survival prediction.

Despite these advancements, several challenges remain unresolved. Whole-slide histopathology images introduce substantial computational complexity due to gigapixel image sizes and expensive preprocessing pipelines. Patch extraction, tissue segmentation, stain normalization, and feature embedding generation require high computational

resources and large-scale storage infrastructure, limiting practical deployment in resource-constrained environments.

The survey also revealed that interpretability remains a major concern in AI-driven healthcare systems. Many deep learning models achieve high predictive accuracy but provide limited biological explanation regarding the reasoning behind predictions. This lack of transparency reduces clinical trust and presents challenges for adoption in real-world healthcare settings.

Pathway-aware multimodal transformer frameworks such as PAMT provide a promising direction for addressing these limitations. By organizing genomic information into biologically meaningful pathway representations and modeling pathway-to-patch interactions using cross-attention mechanisms, such systems improve both predictive capability and interpretability.

Another important discussion point is the challenge of multimodal data alignment. Genomic datasets, clinical survival records, and histopathological images often contain incomplete or inconsistent patient identifiers, making cohort construction a significant challenge in multimodal cancer research. Effective patient matching and cross-modal alignment therefore remain critical components of future research pipelines.

Overall, the literature strongly suggests that future cancer survival prediction systems will increasingly emphasize biologically interpretable multimodal transformer architectures capable of integrating heterogeneous biomedical capabilities within clinically explainable AI frameworks.

5. CONCLUSION

This survey examined recent advancements in artificial intelligence-based cancer survival prediction systems with particular focus on multimodal learning frameworks integrating genomic, histopathological, and clinical information. The review analyzed traditional statistical methods, machine learning approaches, deep learning architectures, transformer-based systems, and pathway-aware multimodal frameworks developed for computational oncology applications.

The survey findings indicate that multimodal approaches generally outperform unimodal systems due to their ability to capture complementary molecular and morphological characteristics associated with tumor progression and patient prognosis. Recent transformer-based architectures demonstrated improved capability in modeling long-range dependencies, cross-modal interactions, and complex biological relationships compared to

conventional machine learning and convolutional neural network-based approaches.

The comparative analysis also revealed several important limitations in existing systems, including insufficient interpretability, coarse multimodal fusion mechanisms, computational inefficiency in whole-slide image processing, and inadequate genotype-phenotype interaction modeling. These challenges limit the clinical applicability and scalability of many current survival prediction frameworks.

Among the surveyed approaches, pathway-aware multimodal transformer architectures such as PAMT emerged as a promising direction for interpretable cancer survival analysis. By integrating pathway-level genomic representations with histopathological image features using attention-based multimodal interaction mechanisms, such systems improve both predictive performance and biological contextualization.

Overall, the survey highlights the growing importance of biologically interpretable multimodal AI systems in computational oncology. Future research is expected to focus on scalable multimodal fusion frameworks, improved cross-modal alignment techniques, efficient whole-slide image processing, and clinically explainable transformer-based survival prediction systems capable of supporting precision medicine applications.

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